

GridPulse Insights - Founder / AI & Energy Systems

Grid Intelligence | Physics-Based Modelling | AI Validation | Product & Venture Building

PROFILE

Energy-systems technologist and founder building GridPulse Insights, a platform that helps data-centre developers identify locations where grid connection is most feasible before committing to lengthy and costly formal studies. Combines an M.Sc. in Power Engineering, an M.Sc. in Management and more than five years of experience across energy-market modelling, forecasting, backend systems, cloud delivery and production support. Builds with a clear evidence boundary: public-data screening and uncertainty-aware analysis support better decisions but do not replace operator-confirmed capacity.

FOUNDER & TECHNICAL STRENGTHS

Energy & grid	Power systems; electricity markets; grid constraints; renewable integration; forecasting; scenario analysis
AI & modelling	Time series; regression; probabilistic reasoning; model validation; physics-constrained analysis; uncertainty communication
Product & data	Python; C#.NET; SQL; PostgreSQL; APIs; data pipelines; geospatial workflows; analytical interfaces
Execution	Customer discovery; technical requirements; testing; CI/CD; cloud delivery; production readiness; stakeholder alignment

VENTURE WORK

- Founder / Product Builder | GridPulse Insights, Berlin

Current

AI-assisted grid-capacity screening for data-centre site selection

 - Built a working Power Finder and backtesting environment that combines public German grid, generation, storage and infrastructure evidence for candidate-site screening.
 - Designed a capacity-envelope methodology using scenario-based electrical analysis, explicit assumptions, provenance, confidence and a clear distinction between screening results and DSO-confirmed capacity.
 - Developed the product across data ingestion, backend services, modelling, map-based investigation and decision-ready outputs while continuously auditing for duplicate or unsupported logic.
 - Defined the next validation path: benchmark against reference networks, test model uncertainty, secure expert review and run a pilot with a data-centre or grid-facing partner.
- Product Builder | RegenSync, Berlin

Earlier venture

Workflow platform for navigating regenerative-therapy pipelines

 - Explored a role-aware platform for navigating complex regenerative-therapy evidence and operational pathways.
 - Learned to narrow the first customer problem before expanding platform scope, a lesson now applied directly to GridPulse's focused market entry.

PROFESSIONAL EXPERIENCE

- Energy Market Consultant | Baringa Management, Dusseldorf

11/2024 - 05/2026

Energy technology delivery, production readiness and cross-functional ownership

 - Supported an international energy-IT transformation across trading, business, development, testing and client teams.
 - Owned defined workstreams and structured technical requirements, dependencies, risks, milestones and delivery decisions.
 - Coordinated testing, defect resolution, production cutover and stabilisation across technical and operational stakeholders.
- Full Stack DevOps / C# Developer | Xapio, Munich

11/2022 - 09/2024

Backend services, relational data workflows, cloud delivery and software quality

 - Developed modular C#.NET and Blazor applications, backend services and PostgreSQL-based data workflows.
 - Worked with Azure Functions, App Services, Git, CI/CD, xUnit testing and controlled deployment processes.

PROFESSIONAL EXPERIENCE CONTINUED

Quantitative Analyst - Energy Markets | GETEC Energie, Hannover

06/2021 - 10/2022

Python forecasting, energy-market modelling and user-facing analytical tools

- Developed Python short-term and day-ahead electricity-price forecasts and gas-price models for TTF and THE.
- Processed and validated time-series data, market fundamentals, scenarios and model assumptions.
- Created a user-oriented Python interface and Power BI reports that made forecast outputs accessible to analysts.
- Supported PPA optimisation and trading-strategy analysis by converting quantitative results into practical decisions.

Quantitative Analyst for Market Models | Likron, Munich

11/2019 - 05/2021

Algorithmic-trading software, Python prototypes, AWS data workflows and support

- Developed Python prototypes for model calibration and C# components for energy-trading and execution systems.
- Built SQL- and AWS-based market-data workflows and automation supporting monitoring and operational reporting.
- Tested software behaviour, investigated data issues and supported live trading applications.

Quantitative Analysis Intern | Statkraft Markets, Dusseldorf

10/2018 - 03/2019

Python time-series modelling, statistical validation and domain collaboration

- Developed UK imbalance-price and Net Imbalance Volume models using Python and regression methods.
- Validated large time-series datasets, calculation logic, model assumptions and forecast behaviour.
- Documented methods and limitations and communicated results to trading and market-analysis specialists.

EDUCATION

M.Sc. Management | TUM School of Management

2022 - 2025

Focus: Entrepreneurship, business creation, technology assessment and product-oriented decision-making

M.Sc. Power Engineering | Technical University of Munich

2016 - 2019

Focus: Power systems, electricity markets, statistics, renewable integration and quantitative energy modelling

B.Sc. Mechanical Engineering | Islamic Azad University Mashhad

2010 - 2014

Focus: Renewable energy, energy conversion, thermodynamics and engineering fundamentals

TECHNICAL TOOLKIT

Python | C#.NET | SQL | PostgreSQL | AWS | Azure | Git | CI/CD | xUnit | Power BI | Excel/VBA | Blazor | Data pipelines | Forecasting | Model validation

LANGUAGES & LOCATION

English C1 | German B2 | Farsi Native | German and EU citizen | Based in Berlin

BUILDER PROFILE

I build at the intersection of energy systems, AI and production software. GridPulse began as an unanswered infrastructure question and became a working product because I prefer testing assumptions through implementation, evidence and user feedback.

FIT FOR TUM.AI E-LAB 6.0

- Built GridPulse independently across public-data ingestion, backend services, grid-context analysis, capacity-envelope backtesting and decision-ready outputs.
- Combines AI inference and forecasting with power-system physics, explicit uncertainty and abstention when the available evidence cannot support a reliable claim.
- Brings both technical and commercial foundations through an M.Sc. in Power Engineering and an M.Sc. in Management from TUM.
- Contributes practical experience from energy forecasting, trading software, cloud delivery, testing and cross-functional product implementation.
- Wants to use the E-Lab to challenge the AI architecture, validate customer urgency, sharpen the first marketable product and secure a pilot-ready path.

WHAT I CONTRIBUTE

Energy-domain depth, hands-on product building, honest communication of model limitations, and the willingness to help other teams translate ambitious AI ideas into testable systems.